

07/28/26

Hardware Requirements

Tier	REQ ID	Parent ID	Requirement Text	Verification
1	PFD-HW-001	-	The PFD Display Unit shall provide hardware computing, supervisory control, aircraft-data interface, memory, storage, display, power-management, reset, watchdog, and maintenance.	Inspection
2	PFD-HW-100	PFD-HW-001	The PFD Display Unit shall provide a Linux application and graphics processing function.	Test
3	PFD-HW-101	PFD-HW-100	The Linux application and graphics processing function shall use one AMD XQZU3EG-1SFRA484I XQ Zynq UltraScale+ MPSoC.	Inspection
3	PFD-HW-102	PFD-HW-100	The application processor hardware shall provide interfaces for DDR4 memory, eMMC storage, QSPI boot flash, and DisplayPort-class display output as supported by the selected MPSoC.	Inspection + Test
3	PFD-HW-103	PFD-HW-100	The MPSoC shall be installed in the 484-ball FCBGA package defined by the selected component.	Inspection
2	PFD-HW-110	PFD-HW-001	The PFD Display Unit shall provide an independent RTOS/supervisory processing function.	Test
3	PFD-HW-111	PFD-HW-110	The supervisory processing function shall use one Texas Instruments S5LS20216ASPGEMEP TMS570LS20216-EP Hercules MCU.	Inspection
3	PFD-HW-112	PFD-HW-110	The supervisory MCU implementation shall provide SPI, UART, and JTAG connectivity required by the PFD hardware architecture.	Inspection + Test
3	PFD-HW-113	PFD-HW-110	The supervisory MCU shall be electrically independent of the Linux application processor such that supervisory functions are not dependent on Linux execution.	Inspection + Test
2	PFD-HW-120	PFD-HW-001	The PFD Display Unit shall provide ARINC 429 aircraft-data communications.	Test
3	PFD-HW-121	PFD-HW-120	The ARINC 429 interface shall use two Holt HI-35930PCTF devices.	Inspection
3	PFD-HW-122	PFD-HW-120	The two HI-35930PCTF devices shall provide an aggregate capability of four ARINC 429 receive channels and two ARINC 429 transmit channels.	Inspection + Test
3	PFD-HW-123	PFD-HW-120	The ARINC 429 devices shall communicate with the supervisory MCU through SPI.	Inspection + Test
3	PFD-HW-124	PFD-HW-120	The external aircraft-data connector implementation shall route the required ARINC 429 differential signals between the PFD and aircraft wiring.	Inspection + Continuity Test

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2	PFD-HW-130	PFD-HW-001	The PFD Display Unit shall provide DDR4 system memory for the MPSoC.	Test
3	PFD-HW-131	PFD-HW-130	The DDR4 memory implementation shall use two ISSI IS46QR16512A-075VBLA2 devices, each rated as 8-Gbit x16 DDR4 SDRAM.	Inspection
3	PFD-HW-132	PFD-HW-130	The two DDR4 devices shall provide a combined 16-Gbit memory capacity and a combined x32 data-width implementation.	Inspection + Test
2	PFD-HW-140	PFD-HW-001	The PFD Display Unit shall provide nonvolatile boot/configuration storage.	Test
3	PFD-HW-141	PFD-HW-140	Boot/configuration storage shall use one Micron MT25QL512ABB8E12-0AUT 512-Mbit Quad-SPI NOR flash device.	Inspection
3	PFD-HW-142	PFD-HW-140	The QSPI boot-flash interface shall operate from the 3.3 V interface requirement identified for the selected flash component.	Inspection + Test
2	PFD-HW-150	PFD-HW-001	The PFD Display Unit shall provide main nonvolatile storage for the Linux processing environment.	Test
3	PFD-HW-151	PFD-HW-150	Main nonvolatile storage shall use one Swissbit SFEM020GB2ED1TB-A-CE-11P-STD 20 GB eMMC device.	Inspection
3	PFD-HW-152	PFD-HW-150	The eMMC device shall interface with the MPSoC through the eMMC interface.	Inspection + Test
2	PFD-HW-160	PFD-HW-001	The PFD Display Unit shall provide controlled multi-rail power for the MPSoC and associated processor rails.	Test
3	PFD-HW-161	PFD-HW-160	The processor power-management function shall use one Texas Instruments TPS65086401RSKT multi-rail PMIC.	Inspection
3	PFD-HW-162	PFD-HW-160	The PMIC shall provide I2C-accessible power-management control to the PFD electronics.	Inspection + Test
3	PFD-HW-163	PFD-HW-160	The PFD design shall include the external passive and support components required to implement the PMIC power rails defined by the final schematic.	Inspection
2	PFD-HW-170	PFD-HW-001	The PFD Display Unit shall provide hardware reset supervision.	Test
3	PFD-HW-171	PFD-HW-170	Hardware reset supervision shall use one Texas Instruments TPS3808G01MDBVTEP adjustable power/reset supervisor.	Inspection
3	PFD-HW-172	PFD-HW-170	The reset supervisor shall provide a hardware reset output to the supervised PFD processing circuitry when the configured monitored condition is not satisfied.	Test
2	PFD-HW-180	PFD-HW-001	The PFD Display Unit shall provide an independent hardware watchdog function.	Test
3	PFD-HW-181	PFD-HW-180	The independent watchdog function shall use one Texas Instruments TPS3431SDRBR window watchdog IC.	Inspection

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3	PFD-HW-182	PFD-HW-180	The watchdog shall detect processor servicing outside the configured watchdog window and provide the associated hardware fault response defined by the final schematic.	Test
2	PFD-HW-190	PFD-HW-001	The PFD Display Unit shall provide a dedicated primary-flight display panel.	Demonstration
3	PFD-HW-191	PFD-HW-190	The PFD display panel shall use the YOURITECH ET140FH01-HG 14-inch LCD selected in the finalized BOM.	Inspection
3	PFD-HW-192	PFD-HW-190	The PFD display panel shall provide a native resolution of 1920 x 1080 pixels.	Inspection + Test
3	PFD-HW-193	PFD-HW-190	The PFD display panel shall provide a specified luminance of 1000 nits.	Inspection + Test
3	PFD-HW-194	PFD-HW-190	The PFD display panel shall interface to the PFD electronics through eDP.	Inspection + Test
2	PFD-HW-200	PFD-HW-001	The PFD Display Unit shall provide the reference clocks and oscillators required by the selected MPSoC.	Inspection + Test
3	PFD-HW-201	PFD-HW-200	Final oscillator frequencies and component values shall be selected to satisfy the MPSoC clock-input requirements used by the final schematic.	Analysis + Inspection
2	PFD-HW-210	PFD-HW-001	The PFD Display Unit shall provide a JTAG hardware-debug/programming interface.	Inspection + Demonstration
3	PFD-HW-211	PFD-HW-210	The JTAG interface shall include the connector and support circuitry required to access the applicable PFD processing devices during development and maintenance.	Inspection
2	PFD-HW-220	PFD-HW-001	The PFD Display Unit shall provide a UART maintenance interface.	Demonstration
3	PFD-HW-221	PFD-HW-220	The UART maintenance interface shall include the connector and level-interface circuitry required by the final electrical design.	Inspection + Test
2	PFD-HW-230	PFD-HW-001	The PFD Display Unit shall provide the display connector and backlight-interface support required by the ET140FH01-HG LCD.	Inspection + Test
2	PFD-HW-240	PFD-HW-001	The PFD Display Unit shall provide a rugged aircraft external I/O connector for aircraft data and unit electrical interfaces.	Inspection
3	PFD-HW-241	PFD-HW-240	The final aircraft external connector shall provide sufficient contacts for the finalized PFD aircraft I/O and power wiring defined by the schematic.	Inspection + Continuity Test
2	PFD-HW-250	PFD-HW-001	The PFD Display Unit shall include thermal-management hardware for heat transfer from the processing electronics.	Inspection + Thermal Test
3	PFD-HW-251	PFD-HW-250	Thermal-management hardware shall include the heat spreader, heatsink, and/or thermal-interface material required by the final mechanical and thermal analysis.	Analysis + Inspection

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2	PFD-HW-260	PFD-HW-001	The PFD Display Unit shall include the decoupling, bias, pull-up/pull-down, and termination networks required by the SoC, memory, ARINC 429, and PMIC circuits.	Inspection
3	PFD-HW-261	PFD-HW-260	Final passive values and quantities shall be established from the selected-component datasheets, reference designs, and completed schematics.	Analysis + Inspection